ARTIFICIAL INTELLIGENCE

will redefine security risk management

The most influential trend, 2026–2031

THREAT LANDSCAPE + DEFENCE + GOVERNANCE

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Why AI is the leading contender

Survey findings illustrate the breadth of expected change.

94%

Expect AI to be the most significant driver of cybersecurity change in the coming year.

87%

Identify AI-related vulnerabilities as the fastest-growing cyber risk in 2025.

AI changes three things at once

ATTACK METHODS • **DEFENSIVE CAPABILITIES** • **RISK GOVERNANCE**

Source: World Economic Forum, Global Cybersecurity Outlook 2026. Respondents' perceptions, not forecasts guaranteed to occur.

The new threat landscape

Attackers can use AI, while poorly secured AI becomes a target.

THREAT	VULNERABILITY	BUSINESS IMPACT
AI-enhanced phishing	Human trust and weak authentication	Fraud / credential theft
Automated vulnerability research	Unpatched software and misconfiguration	System compromise
Deepfake impersonation	Weak identity verification	Fraud / reputational harm
Prompt injection	Unsafe AI tool access	Data disclosure / unwanted actions
AI data leakage	Weak classification and access controls	Privacy / IP losses

NCSC (2025): AI is already assisting reconnaissance, social engineering and vulnerability research.

From periodic assessment to continuous insight

A human-supervised AI risk management loop.

01

COLLECT

Logs • vulnerabilities • intelligence

→

02

ANALYSE

Detect anomalies • correlate alerts

→

03

DECIDE

Human validation • risk prioritisation

→

04

ACT & LEARN

Contain • remediate • monitor

Critical limitation: predictions can be wrong; high-impact actions need accountable human oversight.

Case study: safeguarding FTTP infrastructure

Hypothetical UK broadband provider: compromised customer equipment and network-management access.

TRADITIONAL SECURITY

- Static detection signatures
- Manual inspection of alerts
- Periodic vulnerability reviews
- Analyst-led initial investigation

AI-ENHANCED SECURITY

- Behavioural anomaly detection
- Automated alert correlation
- Continuous risk prioritisation
- AI-assisted incident triage

Operational safeguard: avoid automated blocking that could disconnect legitimate customers.

Govern AI. Do not replace risk management.

Combine established controls with specialist AI safeguards.

NIST CSF 2.0	Govern • Identify • Protect • Detect • Respond • Recover
NIST AI RMF	Govern • Map • Measure • Manage
ISO/IEC 27001	Risk assessment • treatment • assurance
OWASP LLM Top 10	Prompt injection • information leakage and other LLM risks
SIEM / SOAR	Monitoring • alert triage • controlled response

Additional safeguards: least privilege • testing • audit trails • human approval • periodic revalidation

The difficult questions: ethics, law and trust

Technical effectiveness is not enough.

LEGAL	UK GDPR; Data Protection Act 2018; lawful processing
ETHICAL	Bias; explainability; transparent accountability
SOCIAL	Workplace surveillance; jobs; public confidence
PROFESSIONAL	Competence; confidentiality; independent judgement
OPERATIONAL	False positives; manipulation; overreliance

Why AI rather than another trend?

Assess influence across the whole SRM lifecycle.

Quantum computing	Potentially serious future encryption challenge	AI already affects many current security activities
Zero Trust	Limits implicit network trust	AI can support adaptive risk-based decisions
Cloud security	Protects critical distributed systems	AI cuts across cloud attacks and defence
Ransomware	Severe operational and financial damage	AI reshapes both attacker and defender capability

Five-year outlook: 2026–2031

Scenario-based forecasts, not guaranteed outcomes.

2026–27

AI-assisted detection and SOC workflows

2027–28

Stronger AI governance and validation

2028–29

Human-supervised agentic investigations

2029–30

Cross-sector AI risk analytics

2030–31

More mature controlled autonomous security

Dependencies: technical maturity • model reliability • economic incentives • regulatory decisions

WHY AI WINS

01 **Reshapes cyberattacks and defence**

02 Changes the speed and scope of risk assessment

03 Influences almost every digital sector

04 Creates new governance, legal and ethical risks

05 Demands continuous evidence-based oversight

The future is not AI replacing judgement. It is AI with accountable human judgement.

References and further reading

Selected authoritative sources • Harvard style

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Accessed: 17 October 2026 | Slide 11 is an appendix and is not included in the 10-minute speaking allocation.